

React Introduction Session

What is React?

React is an **open-source JavaScript library** used to build **fast, interactive, and reusable User Interfaces (UI)** for web applications. It was developed by **Facebook (Meta)** in **2013** to solve the problem of updating large and dynamic web pages efficiently. Unlike HTML, CSS, and JavaScript alone, React helps developers break a webpage into small reusable pieces called **Components**.

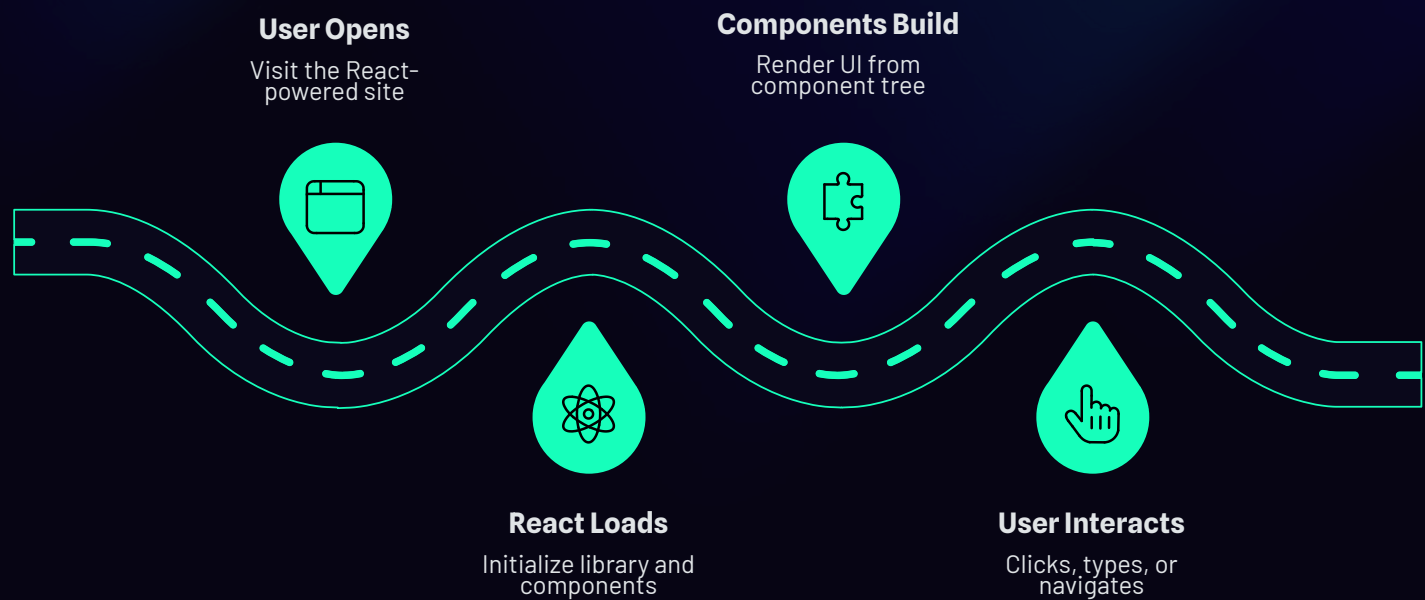
Why was React created?

Before React, developers had to manually update different parts of a webpage using JavaScript. As applications became larger, maintaining the code became difficult. React solved this problem by updating only the necessary part of the page.

Real World Applications

Facebook	Instagram	Netflix
WhatsApp Web	Airbnb	Swiggy
Zomato		

Flowchart



Key Points

1 ✓ JavaScript Library	2 ✓ Developed by Meta	3 ✓ Component Based	Made with GAMMA
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Why React?

Problem with Traditional JavaScript

Suppose you're building an Ecommerce website.
The page contains

- Navbar
- Banner
- Product List
- Cart
- Footer

If the user adds one product to the cart,
Traditional JavaScript may update many DOM elements manually.

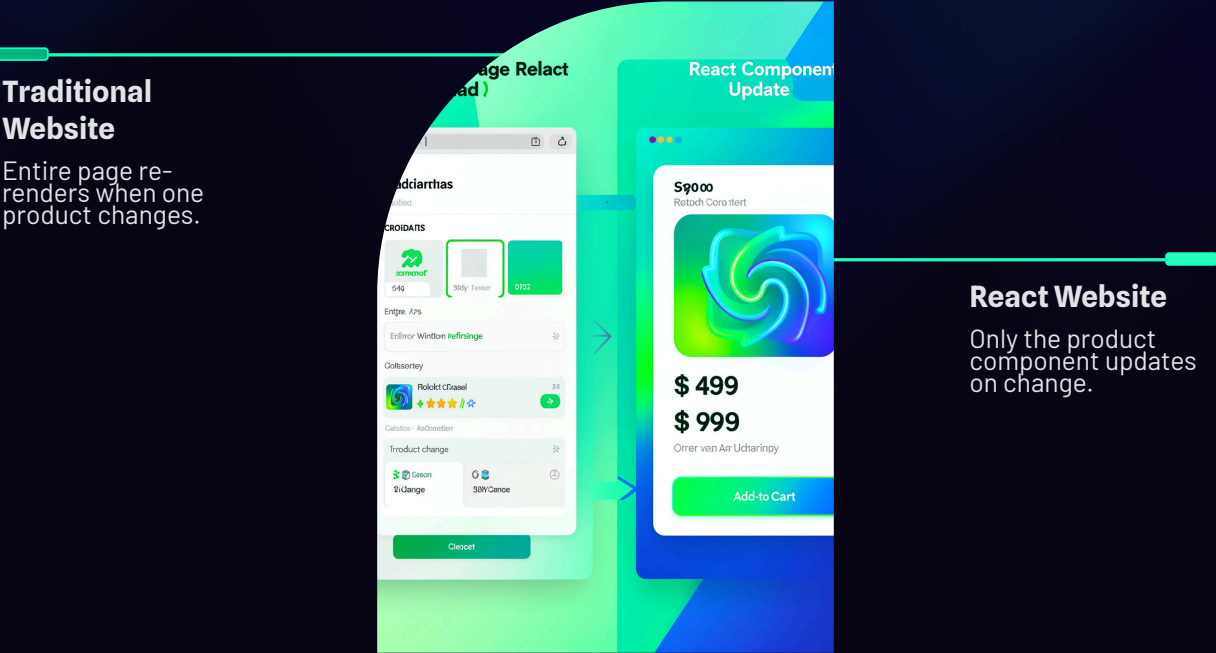
As applications grow,

- Code becomes lengthy.
- Maintenance becomes difficult.
- Performance becomes slower.

React Solution

React divides the webpage into independent components.
Each component manages its own data.
Only the affected component updates.

Flowchart



Advantages

Faster Applications

Reusable Components

Less Code

Better Performance

Made with GAMMA

SPA vs MPA

Multi Page Application (MPA)

Every click requests a completely new HTML page from the server.

Examples

- Traditional Banking Websites
- Government Websites
- Older Ecommerce Websites

MPA Flow

User Clicks Link → Browser Sends Request → Server Creates New HTML → Entire Page Reloads

Single Page Application (SPA)

Only data changes.

The webpage remains loaded.

JavaScript updates the UI.

Examples

- Gmail
- Facebook
- Instagram
- React Applications

SPA Flow

User Clicks Link → React Fetches Data → Updates Component → No Page Reload

Comparison

MPA	SPA
Reloads page	No reload
Slower	Faster
More server requests	Fewer server requests
Traditional websites	React applications

React Architecture

How React Works

React follows a Component Based Architecture.
Every webpage is divided into small independent components.

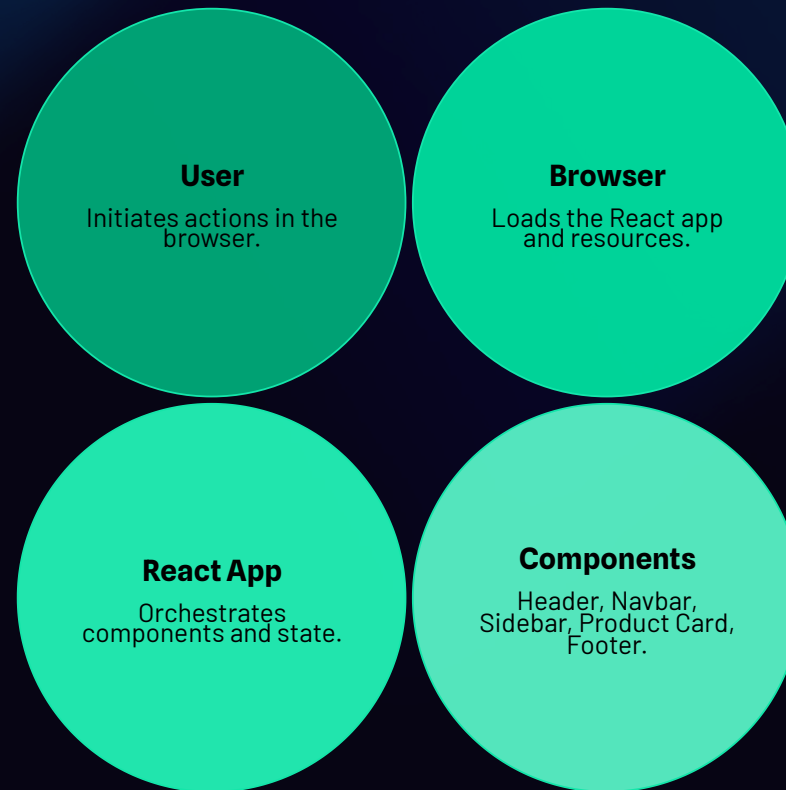
Example

Amazon Homepage

- Header
- Search Bar
- Product Card
- Sidebar
- Footer

Each is a separate component.

Architecture Flow



Benefits

Independent Components

Easy Testing

Reusable Code

Better Team Collaboration

Easy Maintenance

Made with **GAMMA**

Real Life Example

Virtual DOM, Diffing and Reconciliation

What is DOM?

DOM stands for

Document Object Model

The browser converts HTML into a tree structure called DOM.

JavaScript changes webpage content using the DOM.

Problem with Real DOM

Every change updates the browser.

Browser rendering is expensive.

Frequent updates reduce performance.

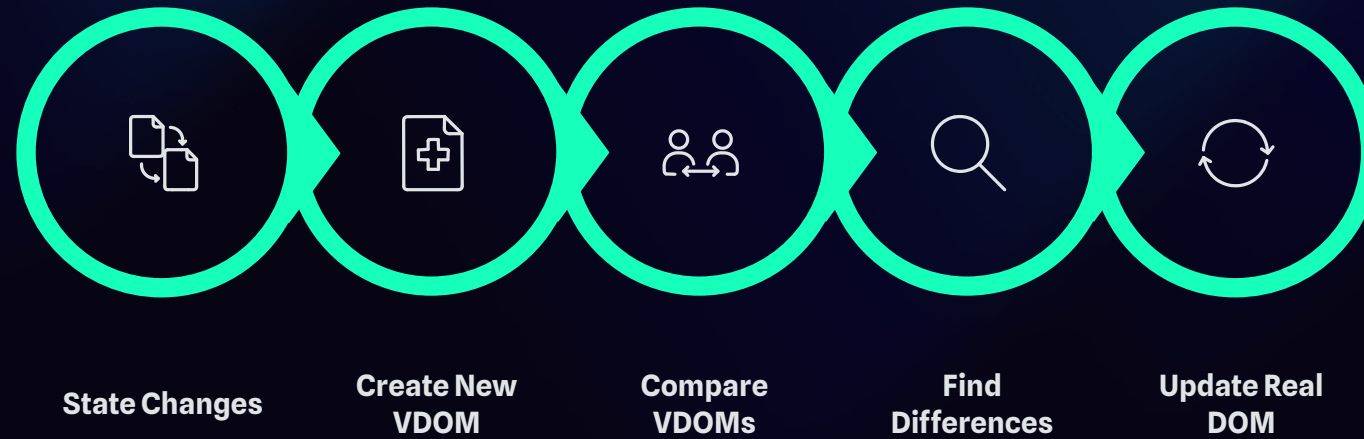
What is Virtual DOM?

Virtual DOM is a lightweight JavaScript copy of the Real DOM.

Whenever data changes,

React first updates the Virtual DOM.

Virtual DOM Flow



What is Diffing?

Diffing is the process of comparing

Old Virtual DOM

with

New Virtual DOM.

React identifies only the changed elements.

Example

Old

Hello

New

Hello Harinath

React detects

Only text changed.

Not the whole page.

What is Reconciliation?

Reconciliation is the process of updating the Real DOM after Diffing.

JSX

What is JSX?

JSX stands for
JavaScript XML.
It allows us to write HTML inside JavaScript.

Without JSX

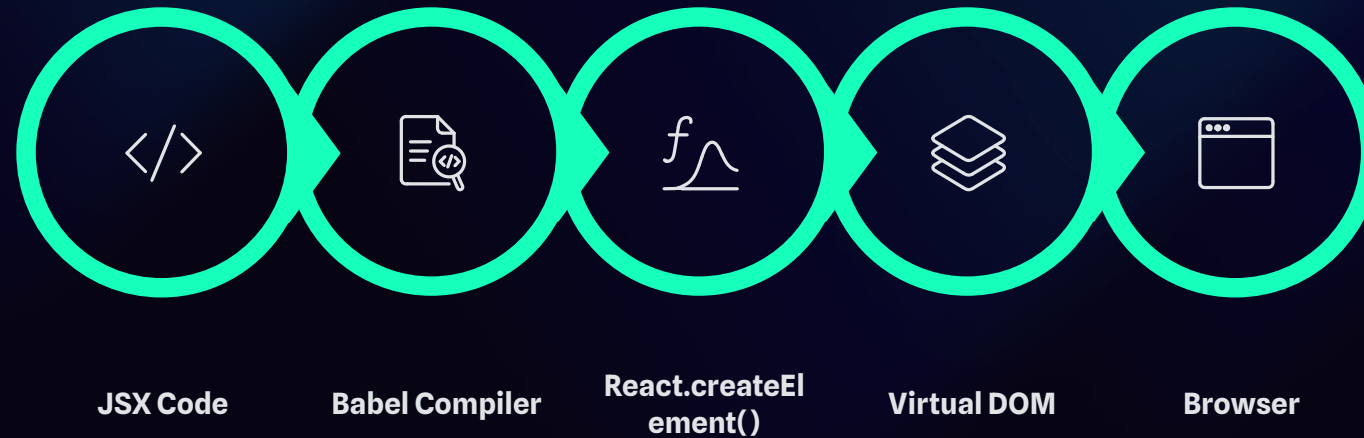
```
React.createElement(  
  "h1",  
  null,  
  "Hello React"  
);
```

With JSX

```
<h1>Hello React</h1>
```

Much easier to read.

JSX Compilation Flow



JSX Rules

One parent element

Use className

Use camelCase

Expressions inside {}

Example

```
const name="Harinath";  
<h1>Hello {name}</h1>
```

React Folder Structure and Vite Setup

Create React Project

```
npm create vite@latest
```

- Choose
- React
 - JavaScript
- Then

```
cd react-app
npm install
npm run dev
```

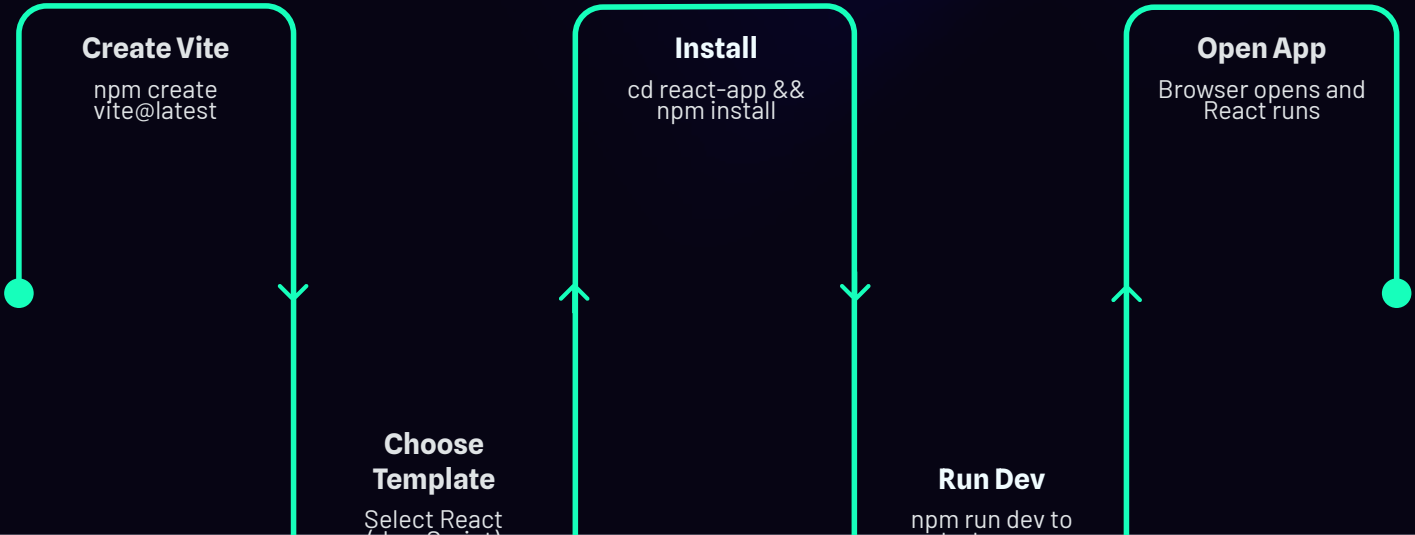
Folder Structure



Folder Explanation

src Application source code.	App.jsx Main component.	main.jsx Application entry point.
assets Images and fonts.	public Static files.	

Vite Startup Flow



Components and Functional Components

What is a Component?

A Component is a reusable part of the UI.
Instead of writing one huge HTML page,
React divides it into small blocks.

Example

Amazon Home Page

- Header
- Navbar
- Banner
- Product Card
- Footer

Each is a separate component.

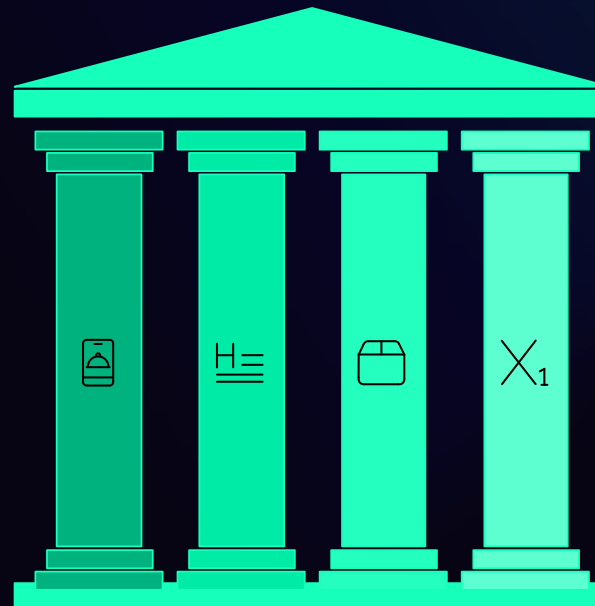
Component Flow

App Root

Holds all page components together

Product Cards

Three reusable product components



Header & Navbar

Top navigation and branding

Footer

Site links and legal info

Functional Component

```
function Header(){  
  return(  
    <h1>Welcome Students</h1>  
  );  
}
```

Advantages

Reusable

Easy Testing

Simple

Made with **GAMMA**

Mini Project

Build Hello React Application

Step 1

Create Project
↓
Run Application
↓
Remove Unnecessary Code
↓
Display
Hello React

Step 2

Create Student Component

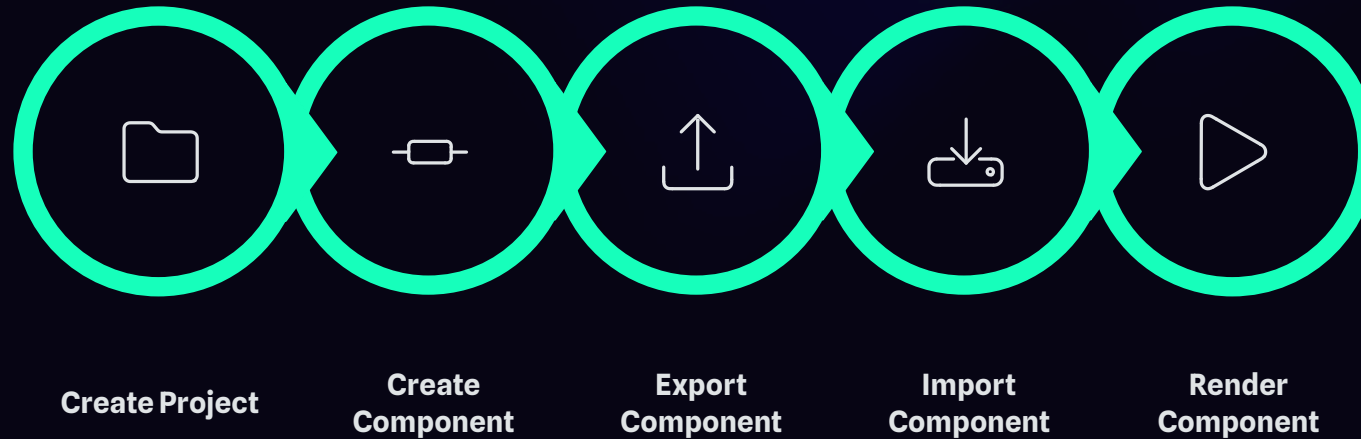
```
function Student(){  
  return(  
    <h2>Harinath</h2>  
    <p>React Developer</p>  
  );  
}  
export default Student;
```

Step 3

Import into App.jsx

```
import Student from "./Student";  
  
function App(){  
  return(  
    <>  
    </>  
  );  
}
```

Project Flow



Summary, Homework and Interview Questions

Day 1 Summary

Today we learned

React	Why React	SPA
MPA	Architecture	Virtual DOM
Diffing	Reconciliation	JSX
Folder Structure	Vite	Components
Functional Components		

Homework

Build these reusable components.

Header	Navbar	Student Card
Button	Footer	

Display all inside App.jsx.

Interview Questions

1. What is React?
2. Why do we use React?
3. SPA vs MPA?
4. Explain React Architecture.
5. What is DOM?
6. What is Virtual DOM?
7. Explain Diffing.
8. Explain Reconciliation.
9. What is JSX?
10. What is Babel?
11. What is a Component?
12. Functional Component vs Class Component?
13. What is Vite?
14. Why is Vite faster than CRA?
15. Explain the React rendering process.